

Palladium-Catalyzed Cross-Coupling Reaction of $\text{AlArEt}_2(\text{THF})$ with Aryl Bromides

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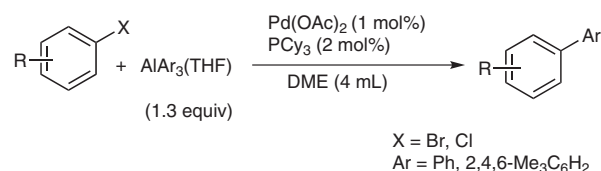
Abstract: Efficient cross-coupling reactions of aryl bromides with $\text{AlArEt}_2(\text{THF})$ ($\text{Ar} = \text{Ph}$, 4- MeC_6H_4 , 4- MeOC_6H_4 , 4- $\text{Me}_3\text{SiC}_6\text{H}_4$, or 2-naphthyl) catalyzed by a catalytic system of 1 mol% $\text{Pd}(\text{OAc})_2$ and 2 mol% PCy_3 furnish aryl–aryl coupling products. The system works well for a wide range of aryl bromides regardless of the electronic or steric nature of the substituent on the aryl bromides.

Key words: cross-coupling, palladium, aryl, aluminum, biaryl

Biaryls are versatile intermediates in organic synthesis and many natural products that possess a biphenyl scaffold display a wide range of biological activity.¹ The palladium-catalyzed cross-coupling reactions represent one of the most widely used processes for the synthesis of biaryls.² Over the past three decades, a tremendous number of catalytic systems using organoborons,³ organotin,⁴ organosilicons,⁵ organozincs or organozirconiums,⁶ and organomagnesiums⁷ as coupling reagents has been established. Recent developments in cross-coupling reactions extended the reaction scope to the inert substrate of aryl chlorides⁸ or phenol derivatives⁹ and to the catalytic systems for the formation of C–N,¹⁰ C–O,¹¹ and C–S bonds.¹² It has been demonstrated that catalytic systems of bulky and electron-rich phosphoranes¹³ and carbenes¹⁴ are highly efficient for coupling reactions of organic halides. In terms of catalytic systems, recent studies centered on developments of catalysts with ligands other than the phosphine- or NHC carbene-based compounds and on systems with metal centers of nickel,¹⁵ copper,¹⁶ or iron.¹⁷ Many catalytic systems have achieved high yields of coupling products. However, the specially designed ligands, which are in general not commercially available, limit those systems for practical applications. In addition to the above-mentioned coupling reagents, a few examples of coupling reactions between aryl halides and alkylaluminum reagents have been reported.¹⁸

Organoaluminum compounds are more reactive than organoboron, organosilicon, and organozinc reagents due to the higher Lewis acidity of the aluminum metal, and it is expected that the organoaluminum reagents, especially arylaluminum compounds, would facilitate coupling reac-

tions with aryl halides in a shorter reaction time.¹⁹ Recently, we have discovered that the arylaluminum compounds were effective reagents in asymmetric aryl additions to aldehydes²⁰ and ketones.²¹ Furthermore, $\text{AlAr}_3(\text{THF})$ compounds were demonstrated to be efficient reagents for coupling reactions with aryl bromides or chlorides (Scheme 1).²² A key advantage of using the $\text{AlAr}_3(\text{THF})$ reagents is that a commercially available and relatively cheap $\text{Pd}(\text{OAc})_2/\text{PCy}_3$ catalytic system is effective enough for aryl–aryl coupling reactions. Another advantage is that an inorganic base as a promoter in many types of coupling reactions is not required. The catalytic system induced coupling reactions of aryl chlorides or bulky aryl bromides under mild reaction conditions at room temperature or $\leq 60^\circ\text{C}$. Importantly, preheating the catalytic system at 100°C for one hour greatly shortened reaction times for couplings of aryl bromides with the most bulky $\text{Al}(2,4,6\text{-Me}_3\text{C}_6\text{H}_2)_3(\text{THF})$ reagent at 40 or 60°C , indicating that the formation of $\text{Pd}(0)$ active species is a rate-determining step while using the bulky arylaluminum reagent.²² However, the use of $\text{AlAr}_3(\text{THF})$ reagents suffers the drawback that only one aryl group in each $\text{AlAr}_3(\text{THF})$ is consumed in the coupling reactions.



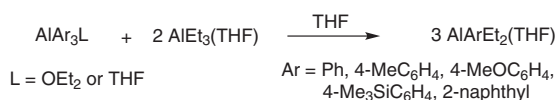
Scheme 1 The cross-coupling reaction of $\text{AlAr}_3(\text{THF})$ with aryl halide

Our recent studies have developed organoaluminum reagents of $\text{Al}(2\text{-furyl})\text{Et}_2(\text{THF})$,²³ $\text{Al}(\text{vinyl})i\text{-Bu}_2(\text{THF})$,²⁴ and $\text{AlArEt}_2(\text{THF})$ ²⁵ for asymmetric additions to ketones. It was found that only products derived from additions of the sp^2 hybridized carbon-based nucleophile, such as 2-furyl, vinyl, or aryl, were obtained. These studies demonstrated that sp^2 hybridized carbon-based nucleophiles were more reactive than the sp^3 hybridized carbon-based alkyl nucleophiles.²⁶ For additions to aldehydes, either there was no observation of alkylation products, or only a minor amount of alkyl addition products were obtained.²⁵ The above studies prompted us to investigate aryl–aryl coupling reactions using $\text{AlArEt}_2(\text{THF})$ reagents.

$\text{AlArEt}_2(\text{THF})$ was indeed an atom-efficient arylaluminum reagent for aryl–aryl cross-coupling reactions, and the cross-coupling reactions of $\text{AlArEt}_2(\text{THF})$ with aryl bromides proceeded smoothly under mild reaction conditions at room temperature over short reaction times of one to four hours except for coupling with the hindered aryl bromides or 3-bromopyridine.

The $\text{AlArEt}_2(\text{THF})$ reagents were easily prepared as colorless oily liquids by mixing one equivalent of AlAr_3L ($\text{L} = \text{OEt}_2$ or THF) and two equivalents of $\text{AlEt}_3(\text{THF})$ in THF at room temperature (Scheme 2) followed by removing the solvent completely. AlEt_3 is a cheap industrial material that can be obtained in bulk, and the above reaction converted one equivalent of $\text{AlAr}_3(\text{THF})$ into three equivalents of $\text{AlArEt}_2(\text{THF})$ reagents.

The cross-coupling reaction between 4-bromoanisole and $\text{AlPhEt}_2(\text{THF})$ using $\text{Pd}(\text{OAc})_2$ and several different phosphine ligands with a $\text{Pd}(\text{OAc})_2/\text{PR}_3$ ratio of 1:2 was first evaluated, and results are listed in Table 1. The results indicate that, without the use of a base, bulky phosphoranes such as PCy_3 and PCy_2Ph were highly active



Scheme 2 The synthesis of $\text{AlArEt}_2(\text{THF})$

ligands to attain the aryl–phenyl coupling product **2a** in high yields of >99% at room temperature over one hour (entries 4 and 5). The PCy_3 ligand was chosen over the PCy_2Ph because of its lower price. Though the $\text{AlPhEt}_2(\text{THF})$ reagent contained both phenyl and ethyl nucleophiles, the coupling reactions produced only the aryl–phenyl coupling product, suggesting that the transmetalation of the phenyl nucleophile was faster than the transmetalation of ethyl groups.

Solvents were then screened. It was observed that the reaction was rather compatible in solvents like toluene, CH_2Cl_2 , THF , and Et_2O , giving product **2a** in excellent yields of 97 to >99% (entries 5, 9–11). However, the reaction did not take place at all in the nonpolar solvent hexane (entry 12). For further optimizations of $\text{Pd}(\text{OAc})_2/\text{PCy}_3$

Biographical Sketches



Wan-Ting Hsu was born in Taichung, Taiwan in 1985. In 2007, she received a B.S. degree from Tunghai University. She then joined the

research group of Prof. Gau as a graduate student at National Chung Hsing University and obtained a Master's degree in June, 2009. Her

research includes aryl–aryl cross couplings of aryl halides and heteroaryl coupling reactions with organoaluminum reagents.



Shuangliu Zhou was born in Anhui, P. R. of China in 1973. He received his B.S. degree from Anhui University of Technology and Science in 1995. He obtained his M.S. and Ph.D. in organ-

ic chemistry from the Anhui Normal University under the supervision of Prof. Shaowu Wang, who focused on synthesis and reactivity of organolanthanide complexes. He was promot-

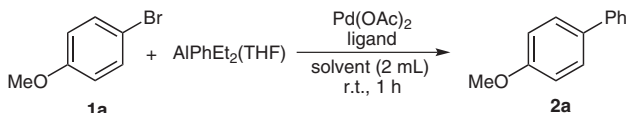
ed to Associate Professor at Anhui Normal University in 2008 and is currently carrying out postdoctoral research with Prof. Han-Mou Gau at National Chung Hsing University.



Han-Mou Gau received his B.S. degree from National Chung Hsing University in 1977. After a two-year military service and two years as a Teaching Assistant at National Chung Hsing University, he went to the Department of Chemistry at Cornell University as a graduate student and re-

ceived his Ph.D. degree in January, 1987. In March, he returned to the Department of Chemistry of National Chung Hsing University as an Associate Professor and was promoted to Professor in 1993. He was the Chair of the Department from 1998 to 2001. His earlier research involved syntheses and

structural characterizations of group 4 metal complexes. Since 1997, he has shifted his research to asymmetric nucleophilic additions to organic carbonyls catalyzed by group 4 metal complexes. Recently, he has also worked on coupling reactions using organoaluminum reagents.

Table 1 Optimizations of Coupling Reaction of 4-Bromoanisole with $\text{AlPhEt}_2(\text{THF})^a$


Entry	Ligand	$\text{Pd}(\text{OAc})_2/\text{PR}_3$ (mol%)	Solvent	$\text{AlPhEt}_2(\text{THF})$ (equiv)	Conv. (%) ^b
1	PPh_3	1:2	toluene	1.3	trace
2	$\text{P}(p\text{-tolyl})_3$	1:2	toluene	1.3	8
3 ^c	tfmpp	1:2	toluene	1.3	trace
4	PCy_2Ph	1:2	toluene	1.3	>99
5	PCy_3	1:2	toluene	1.3	>99
6	dppm	1:2	toluene	1.3	trace
7	dppe	1:2	toluene	1.3	trace
8	dppp	1:2	toluene	1.3	trace
9	PCy_3	1:2	CH_2Cl_2	1.3	97
10	PCy_3	1:2	THF	1.3	>99
11	PCy_3	1:2	Et_2O	1.3	97
12	PCy_3	1:2	hexane	1.3	trace
13	PCy_3	1:1	toluene	1.3	93
14	PCy_3	1:3	toluene	1.3	trace
15	PCy_3	1:2	toluene	1.1	91
16	PCy_3	1:2	toluene	1.5	>99

^a Reaction conditions: 4-bromoanisole (1.0 mmol), solvent (2 mL), r.t., 1 h.

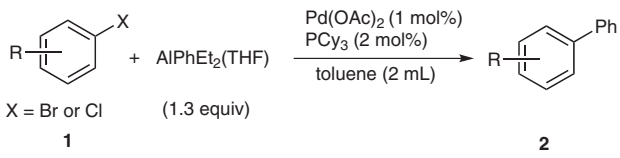
^b Conversions were determined by ^1H NMR spectra.

^c tfmpp: *p*-trifluoromethylphenylphosphine.

ratios, toluene was selected as the solvent. While tuning the ratio to 1:1 (entry 13), the cross-coupling reaction gave product **2a** in a somewhat lower 93% yield. However, increasing the $\text{Pd}(\text{OAc})_2/\text{PCy}_3$ ratio to 1:3 shut down the cross-coupling reaction (entry 14). Subsequently, we studied the effects of the amount of $\text{AlPhEt}_2(\text{THF})$ on the reaction efficiency and found that 1.3 equivalents of $\text{AlPhEt}_2(\text{THF})$ was the optimized quantity (entries 5, 15, and 16) for the coupling reactions.

With the optimized conditions in hand, the scope of the reaction was extended to various aryl bromides with a variety of steric requirements and with substituent groups having a different electronic nature (Table 2). It was found that the catalytic system worked well for a wide range of aryl bromides, regardless of the electronic nature or the steric effect of the substituents on the aryl groups. For example, yields of coupling products ranged from 80 to 86% (entries 1–5) for electron-rich substrates such as 4-methoxybromobenzene, 4-bromothioanisole, 4-methylbromobenzene, 2-methylbromobenzene, and 3,5-dimeth-

ylbromobenzene. For aryl bromides bearing an electron-withdrawing substituent such as 4-fluorobromobenzene, 4-chlorobromobenzene, 4-bromobenzonitrile, and 4-nitrobromobenzene, coupling products were obtained in 72 to 84% yields (entries 8–11). For sterically hindered 2,4-dimethylbromobenzene (entry 6), 1-bromonaphthalene (entry 12), 2-bromonaphthalene (entry 13), coupling products were obtained in high yields of 87 to 95% at room temperature or 60 °C. The system also worked well for the most sterically hindered 2,6-dimethylbromobenzene, affording product **2g** in a satisfactory 89% yield over a longer reaction time of 12 hours at 60 °C (entry 7). It is worth noting that the heterocyclic bromide of 3-bromopyridine was tolerated in this system, furnishing the coupling product **2n** in a satisfactory 80% yield at 40 °C over four hours (entry 14). In contrast, a much lower 14% yield of **2o** (entry 15) was obtained from a coupling reaction with 2-bromopyridine.

Table 2 Coupling of Aryl Halides with $\text{AlPhEt}_2(\text{THF})^a$


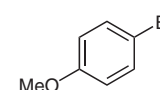
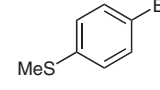
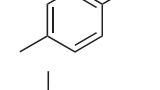
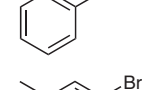
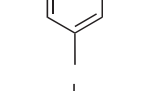
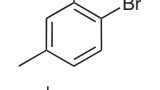
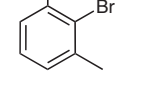
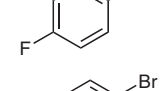
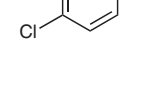
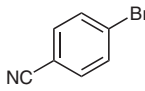
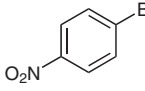
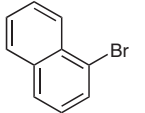
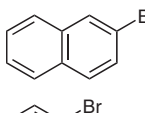
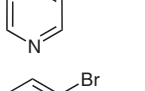
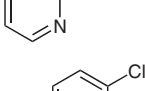
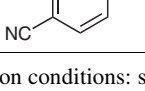
Entry	Substrate	Temp (°C)	Time (h)	Product	Yield (%) ^b
1		r.t.	1	2a	86
2		40	3	2b	86
3		r.t.	1	2c	83
4		r.t.	4	2d	80
5		r.t.	1	2e	81
6		r.t.	3	2f	87
7		60	12	2g	89
8		r.t.	1	2h	82
9		40	2	2i	74

Table 2 Coupling of Aryl Halides with $\text{AlPhEt}_2(\text{THF})^a$ (continued)

$\text{R}-\text{C}_6\text{H}_4-\text{X} + \text{AlPhEt}_2(\text{THF}) \xrightarrow[\text{toluene (2 mL)}]{\text{Pd(OAc)}_2 (1 \text{ mol}\%), \text{PCy}_3 (2 \text{ mol}\%)} \text{R}-\text{C}_6\text{H}_4-\text{Ph}$					
X = Br or Cl (1.3 equiv)					
Entry	Substrate	Temp (°C)	Time (h)	Product	Yield (%) ^b
10		r.t.	3	2j	84
11		r.t.	3	2k	72
12		r.t.	1	2l	95
13		r.t.	1	2m	94
14		40	4	2n	80
15		40	4	2o	14
16		60	12	2j	<5 ^c

^a Reaction conditions: substrate/ $\text{AlEt}_2\text{Ph}(\text{THF})/\text{Pd}(\text{OAc})_2/\text{PCy}_3 = 1.0:1.3:0.010:0.020$ mmol, toluene (2 mL).

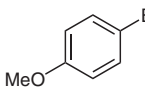
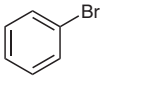
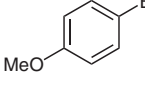
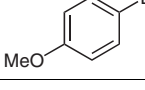
^b Isolated yields are average values of two runs.

^c Conversion was determined by ^1H MNR spectra.

The coupling reaction of 4-cyanochlorobenzene (**1j'**) with $\text{AlPhEt}_2(\text{THF})$ was also investigated. It was expected that the strong electron-withdrawing CN substituent in **1j'** would facilitate the oxidative addition of the inert aryl chloride to the $\text{Pd}(0)$ species. However, only a trace amount of **2j** (<5% conversion) was observed in the coupling reaction carried out at 60 °C over 12 hours (entry 16). Since coupling reactions of aryl bromides with $\text{AlPhEt}_2(\text{THF})$ proceeded smoothly in general at room temperature, the reduction of $\text{Pd}(\text{II})$ to the active $\text{Pd}(0)$ species was not a key factor accounting for the slow reaction of the aryl chloride. The slower reaction reflected the nature of a slower oxidative addition of aryl chlorides. But, **2j** was obtained in an excellent 93% yield in the coupling reaction of **1j'** with $\text{AlPh}_3(\text{THF})$ conducted at the same 60 °C over 12 hours in DME.²² Thus, in addition to the slow oxidative addition of aryl chlorides, the above result strongly suggested that the transmetalation between the $\text{Pd}-\text{Cl}$ bond and the aryl nucleophile of $\text{AlArEt}_2(\text{THF})$ was slower than the transmetalation from $\text{AlAr}_3(\text{THF})$.

Coupling reactions of aryl bromides with different aryl-aluminum compounds, such as $\text{Al}(4\text{-MeC}_6\text{H}_4)_2(\text{THF})$, $\text{Al}(4\text{-MeOC}_6\text{H}_4)_2(\text{THF})$, $\text{Al}(4\text{-Me}_3\text{SiC}_6\text{H}_4)_2(\text{THF})$, or $\text{Al}(2\text{-naphthyl})_2(\text{THF})$ were also studied, and results are summarized in Table 3. Coupling products were obtained in 76 to 90% yields at room temperature over reaction times of one hour (entries 1–4).

Table 3 Coupling of Aryl Bromides with $\text{AlArEt}_2(\text{THF})^a$

$\text{R}-\text{C}_6\text{H}_4-\text{Br} + \text{AlArEt}_2(\text{THF}) \xrightarrow[\text{toluene (2 mL), r.t., 1 h}]{\text{Pd(OAc)}_2 (1 \text{ mol}\%), \text{PCy}_3 (2 \text{ mol}\%)} \text{R}-\text{C}_6\text{H}_4-\text{Ar}$					
(1.3 equiv)					
Entry	Substrate	Ar	Product	Yield (%) ^b	
1		4-MeC ₆ H ₄	2p	81	
2		4-MeOC ₆ H ₄	2a	83	
3		4-Me ₃ SiC ₆ H ₄	2q	76	
4		2-naphthyl	2r	90	

^a Reaction conditions: substrate/ $\text{AlArEt}_2(\text{THF})/\text{Pd}(\text{OAc})_2/\text{PCy}_3 = 1.0:1.3:0.010:0.020$ mmol, toluene (2 mL).

^b Isolated yields are average values of two runs.

In summary, cross-coupling reactions of aryl bromides with $\text{AlArEt}_2(\text{THF})$ ($\text{Ar} = \text{Ph}$, 4-MeC₆H₄, 4-MeOC₆H₄, 4-Me₃SiC₆H₄, or 2-naphthyl) catalyzed by the simple and economic catalytic system of 1 mol% $\text{Pd}(\text{OAc})_2$ and 2 mol% PCy_3 are reported. The reactions afford only aryl-aryl coupling products, indicating that the transmetalation of the aryl nucleophile is faster than that of the ethyl groups. The system is effective enough for a wide range of aryl bromides regardless of the electronic or steric nature of the substituent on the aryl bromides. However, the coupling of $\text{AlPhEt}_2(\text{THF})$ with 4-cyanochlorobenzene affords only a trace amount of the coupling product. An advantage of the $\text{AlArEt}_2(\text{THF})$ reagents is that the atom-efficiency of the $\text{AlArEt}_2(\text{THF})$ reagents is three times that of the $\text{AlAr}_3(\text{THF})$ reagents in terms of the aryl group consumed. However, for coupling reactions of inert aryl chlorides, a slower transmetalation of the aryl nucleophile from $\text{AlArEt}_2(\text{THF})$ to the $\text{Pd}(\text{II})$ species than the transmetalation from $\text{AlAr}_3(\text{THF})$ is suggested. Further studies of coupling reactions employing organoaluminum reagents are under way.

All syntheses and manipulations were carried out under dry N_2 using standard Schlenk techniques or in a glove box. Solvents were dried by refluxing for at least 24 h over P_2O_5 (CH_2Cl_2) or sodium/benzophenone (Et_2O , THF, *n*-hexane, or toluene) and were freshly distilled prior to use. ^1H NMR spectra were obtained with a Varian Mercury-400 (400 MHz) spectrometer, and ^{13}C NMR spectra were

recorded with the Varian Mercury-400 (100.70 MHz). ^1H and ^{13}C chemical shifts were measured relative to TMS as the internal reference. $\text{AlArEt}_2(\text{THF})$ compounds were synthesized according to literature procedure.²⁵

Aryl–Aryl Coupling Reactions; General Procedure

$\text{AlArEt}_2(\text{THF})$ (1.30 mmol) was added to a toluene (2 mL) solution of $\text{Pd}(\text{OAc})_2$ (0.0024 g, 0.010 mmol) and PR_3 (0.020 mmol) under dry N_2 . The mixture was stirred for 10 min, and an aryl halide (1.00 mmol) was added. The resulting solution was stirred for a given period of time and quenched with H_2O (10 mL). The solution was then extracted with CH_2Cl_2 (3×30 mL). The combined organic phases were dried (MgSO_4) and concentrated to dryness under reduced pressure. The residue was purified by column chromatography to give the coupling product (Tables 2 and 3).

4-Methoxybiphenyl (2a)²²

Yield: 0.158 g (86%); white solid.

^1H NMR (400 MHz, CDCl_3): δ = 7.57–7.52 (m, 4 H), 7.44–7.40 (m, 2 H), 7.33–7.29 (m, 1 H), 7.00–6.96 (m, 2 H), 3.86 (s, 3 H).

$^{13}\text{C}\{^1\text{H}\}$ NMR (100 MHz, CDCl_3): δ = 159.1, 140.8, 133.7, 128.7, 128.1, 126.7, 126.6, 114.2, 55.3.

4-Phenylthioanisole (2b)²²

Yield: 0.178 g (86%); white solid.

^1H NMR (400 MHz, CDCl_3): δ = 7.58–7.51 (m, 4 H), 7.45–7.41 (m, 2 H), 7.35–7.32 (m, 1 H), 2.52 (s, 3 H).

$^{13}\text{C}\{^1\text{H}\}$ NMR (100 MHz, CDCl_3): δ = 140.5, 138.0, 137.6, 128.8, 127.4, 127.2, 127.0, 126.8, 15.9.

4-Methylbiphenyl (2c)²²

Yield: 0.140 g (83%); white solid.

^1H NMR (400 MHz, CDCl_3): δ = 7.59–7.56 (m, 2 H), 7.50–7.48 (m, 2 H), 7.44–7.40 (m, 2 H), 7.34–7.30 (m, 1 H), 7.26–7.23 (m, 2 H), 2.39 (s, 3 H).

$^{13}\text{C}\{^1\text{H}\}$ NMR (100 MHz, CDCl_3): δ = 141.1, 138.3, 136.9, 129.4, 128.7, 127.1, 126.9, 126.9, 21.0.

2-Methylbiphenyl (2d)²²

Yield: 0.138 g (80%); yellow liquid.

^1H NMR (400 MHz, CDCl_3): δ = 7.43–7.40 (m, 2 H), 7.36–7.31 (m, 3 H), 7.27–7.24 (m, 4 H), 2.28 (s, 3 H).

$^{13}\text{C}\{^1\text{H}\}$ NMR (100 MHz, CDCl_3): δ = 141.9, 135.2, 130.2, 129.7, 129.1, 128.0, 127.2, 127.1, 126.7, 125.7, 20.4.

3,5-Dimethylbiphenyl (2e)²²

Yield: 0.148 g (81%); yellow liquid.

^1H NMR (400 MHz, CDCl_3): δ = 7.59–7.56 (m, 2 H), 7.44–7.40 (m, 2 H), 7.35–7.31 (m, 1 H), 7.21 (s, 2 H), 7.00 (s, 1 H), 2.38 (s, 6 H).

$^{13}\text{C}\{^1\text{H}\}$ NMR (100 MHz, CDCl_3): δ = 141.4, 141.2, 138.1, 128.8, 128.5, 127.1, 127.0, 125.0, 21.3.

2,4-Dimethylbiphenyl (2f)²²

Yield: 0.159 g (87%); colorless liquid.

^1H NMR (400 MHz, CDCl_3): δ = 7.42–7.38 (m, 2 H), 7.35–7.30 (m, 3 H), 7.15–7.06 (m, 2 H), 2.37 (s, 3 H), 2.25 (s, 3 H).

$^{13}\text{C}\{^1\text{H}\}$ NMR (100 MHz, CDCl_3): δ = 141.9, 139.1, 136.8, 135.1, 131.1, 129.7, 129.3, 128.0, 126.6, 126.5, 21.0, 20.3.

2,6-Dimethylbiphenyl (2g)²²

Yield: 0.162 g (89%); colorless liquid.

^1H NMR (400 MHz, CDCl_3): δ = 7.44–7.40 (m, 2 H), 7.36–7.32 (m, 1 H), 7.18–7.10 (m, 5 H), 2.03 (s, 6 H).

$^{13}\text{C}\{^1\text{H}\}$ NMR (100 MHz, CDCl_3): δ = 141.8, 141.1, 136.0, 129.0, 128.4, 127.2, 127.0, 126.6, 20.8.

4-Fluorobiphenyl (2h)²²

Yield: 0.141 g (82%); yellow solid.

^1H NMR (400 MHz, CDCl_3): δ = 7.55–7.51 (m, 4 H), 7.45–7.41 (m, 2 H), 7.36–7.32 (m, 1 H), 7.14–7.10 (m, 2 H).

$^{13}\text{C}\{^1\text{H}\}$ NMR (100 MHz, CDCl_3): δ = 163.7, 161.2, 140.2, 137.3, 128.8, 128.7, 128.6, 127.2, 127.1, 127.0, 115.7, 115.5.

4-Chlorobiphenyl (2i)^{5b}

Yield: 0.140 g (74%); yellow solid.

^1H NMR (400 MHz, CDCl_3): δ = 7.55–7.48 (m, 4 H), 7.45–7.35 (m, 5 H).

$^{13}\text{C}\{^1\text{H}\}$ NMR (100 MHz, CDCl_3): δ = 139.9, 139.6, 133.3, 128.8, 128.3, 127.5, 127.1, 126.9.

4-Cyanobiphenyl (2j)²²

Yield: 0.172 g (84%); white solid.

^1H NMR (400 MHz, CDCl_3): δ = 7.75–7.68 (m, 4 H), 7.60–7.58 (m, 2 H), 7.51–7.41 (m, 3 H).

$^{13}\text{C}\{^1\text{H}\}$ NMR (100 MHz, CDCl_3): δ = 145.6, 139.1, 132.5, 129.0, 128.6, 127.6, 127.1, 118.8, 110.8.

4-Nitrobiphenyl (2k)²²

Yield: 0.143 g (72%); yellow solid.

^1H NMR (400 MHz, CDCl_3): δ = 8.32–8.29 (m, 2 H), 7.76–7.73 (m, 2 H), 7.65–7.62 (m, 2 H), 7.53–7.43 (m, 3 H).

$^{13}\text{C}\{^1\text{H}\}$ NMR (100 MHz, CDCl_3): δ = 147.5, 147.0, 138.6, 129.0, 128.8, 127.7, 127.3, 124.0.

1-Naphthylbenzene (2l)^{5b}

Yield: 0.194 g (95%); yellow liquid.

^1H NMR (400 MHz, CDCl_3): δ = 7.91–7.85 (m, 3 H), 7.54–7.40 (m, 9 H).

$^{13}\text{C}\{^1\text{H}\}$ NMR (100 MHz, CDCl_3): δ = 140.7, 140.2, 133.8, 131.6, 130.0, 128.7, 128.2, 127.6, 127.2, 126.9, 126.0, 125.7, 125.3.

2-Naphthylbenzene (2m)²⁷

Yield: 0.192 g (94%); white solid.

^1H NMR (400 MHz, CDCl_3): δ = 8.05 (s, 1 H), 7.93–7.86 (m, 3 H), 7.77–7.72 (m, 3 H), 7.53–7.47 (m, 4 H), 7.41–7.37 (m, 1 H).

$^{13}\text{C}\{^1\text{H}\}$ NMR (100 MHz, CDCl_3): δ = 141.0, 138.4, 133.6, 132.5, 128.8, 128.4, 128.1, 127.6, 127.3, 127.3, 126.2, 125.8, 125.7, 125.5.

3-Phenylpyridine (2n)²⁸

Yield: 0.124 g (80%); yellow liquid.

^1H NMR (400 MHz, CDCl_3): δ = 8.86–8.85 (m, 1 H), 8.60–8.59 (m, 1 H), 7.89–7.86 (m, 1 H), 7.60–7.57 (m, 2 H), 7.51–7.35 (m, 4 H).

$^{13}\text{C}\{^1\text{H}\}$ NMR (100 MHz, CDCl_3): δ = 148.3, 148.1, 137.6, 136.5, 134.2, 128.9, 127.9, 127.0, 123.4.

2-Phenylpyridine (2o)²⁹

Yield: 0.022 g (14%); yellow liquid.

^1H NMR (400 MHz, CDCl_3): δ = 8.71–8.69 (m, 1 H), 8.00–7.97 (m, 2 H), 7.78–7.72 (m, 2 H), 7.50–7.40 (m, 3 H), 7.26–7.22 (m, 1 H).

$^{13}\text{C}\{^1\text{H}\}$ NMR (100 MHz, CDCl_3): δ = 157.5, 149.6, 139.4, 136.7, 128.9, 128.7, 126.9, 122.1, 120.6.

4-Methoxy-4'-methylbiphenyl (2p)^{5b}

Yield: 0.161 g (81%); white solid.

¹H NMR (400 MHz, CDCl₃): δ = 7.52–7.50 (m, 2 H), 7.46–7.44 (m, 2 H), 7.24–7.22 (m, 2 H), 6.98–6.96 (m, 2 H), 3.85 (s, 3 H), 2.38 (s, 3 H).¹³C{¹H} NMR (100 MHz, CDCl₃): δ = 158.9, 137.9, 136.3, 133.7, 129.4, 127.9, 126.5, 114.1, 55.3, 21.0.**4-Methoxy-4'-trimethylsilylbiphenyl (2q)³⁰**

Yield: 0.195 g (76%); yellow solid.

¹H NMR (400 MHz, CDCl₃): δ = 7.61–7.54 (m, 6 H), 7.00–6.98 (m, 2 H), 3.84 (s, 3 H), 0.32 (s, 9 H).¹³C{¹H} NMR (100 MHz, CDCl₃): δ = 159.2, 141.2, 138.4, 133.8, 133.6, 128.1, 126.0, 114.2, 55.2, –1.1.**4-Methoxy-2-naphthylbenzene (2r)^{9d}**

Yield: 0.211 g (90%); white solid.

¹H NMR (400 MHz, CDCl₃): δ = 7.99 (s, 1 H), 7.91–7.85 (m, 3 H), 7.73–7.65 (m, 3 H), 7.51–7.44 (m, 2 H), 7.05–7.01 (m, 2 H), 3.88 (s, 3 H).¹³C{¹H} NMR (100 MHz, CDCl₃): δ = 159.2, 138.1, 133.7, 133.6, 132.3, 128.4, 128.3, 128.0, 127.6, 126.2, 125.6, 125.4, 125.0, 114.3, 55.3.HRMS (EI): *m/z* calcd for C₁₇H₁₄O [M⁺]: 234.1045; found: 234.1049.**Acknowledgment**

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